EE2016: Microprocessor theory and lab

Tutorial-1 (Solutions)

Q1. Which addressing mode uses operands that are part of the instruction itself?

A) Direct

B) Register

C) Immediate

D) Indirect

Solution:

C) Immediate

Explanation: The operand is embedded directly in the instruction (e.g., `MOV AL, 05H`).

Q2. In `MOV AX, [BX]`, the addressing mode for the source operand is:

A) Register Indirect

B) Direct

C) Immediate

D) Indexed

{BX] indicates the address pointed by the BX register.

Solution:

A) Register Indirect

Explanation: The operand address is stored in a register (`BX` points to the memory location).

Q3. The addressing mode where the operand address is explicitly given in the instruction is:

A) Immediate

B) Direct

C) Register Indirect

D) Implied

Solution:

B) Direct

Explanation: The instruction contains the absolute memory address (e.g., `MOV AX, [2000H]`).

Q4. For the instruction `PUSH AX`, the addressing mode used for the operand `AX` is:

A) Implied

B) Register

C) Direct

D) Immediate

Solution:

B) Register

Explanation: The operand (`AX`) is explicitly specified in a register.

Q5. In indexed addressing mode, the effective address is calculated as:

A) Base Register + Index Register

B) Index Register + Displacement

C) Program Counter + Displacement

D) Stack Pointer + Offset

Solution:

B) Index Register + Displacement

Explanation: EA = Content of Index Register (e.g., SI/DI) + Displacement (e.g., `MOV AX, ARRAY[SI]`).

Q6 Relative addressing mode is primarily used for:

A) Data transfer

B) Branch instructions

C) Arithmetic operations

D) I/O operations

Solution:

B) Branch instructions

Explanation: The target address is calculated relative to the Program Counter

Q7. The addressing mode where the operand is implied by the instruction (e.g., **STC**) is:

A) Immediate
B) Direct
C) Implied
D) Register

The STC instruction sets the carry flag.

Solution:

C) Implied

Explanation: The operand (Carry Flag) is implicitly defined by the instruction.

Q8. For MOV AX, [BP+10], the source operand uses:

A) Base-Indexed
B) Based Relative
C) Indexed
D) Register Indirect

BP is the base pointer register.

Solution:

B) Based Relative

Explanation: EA = Base Register (`BP`) + Displacement (`10`).

Q9. Consider an 8086 system where:

DS = 4000H
BX = 0200H
DI = 0040H

Memory location 40240H contains 1A2BH.

The instruction MOV AX, [BX+DI+10H] is executed. What value is loaded into AX?

A) 0200H
B) 0040H
C) 1A2BH
D) 40240H

Solution:

C) 1A2BH

Effective Address (EA) = BX + DI + 10H = 0200H + 0040H + 10H = 0250H

Q10. In a processor with Base-Register Addressing, if the base register contains 3000H and the instruction specifies a displacement of +200H, what is the effective address? What happens if the base register is incremented to 3002H before instruction execution?

A) 3200H, 3202H

B) 3200H, 3200H

C) 3000H, 3002H

D) 3200H, 3202H with auto-indexing

Solution:

A) 3200H, 3202H

EA = Base Register + Displacement = 3000H + 200H = 3200H

If base register changes to 3002H, new EA = 3002H + 200H = 3202H. Base registers are mutable unless specified as auto-indexed.

Q11. Consider a 32-bit processor where:

EAX = 00001000H

EBX = 00002000H

Memory location 00003000H contains 00004000H

After MOV ECX, [EAX+EBX*2] , what is ECX?
A) 00003000H
B) 00005000H
C) Content of 00005000H
D) Content of 00003000H

Solution:

C) Content of 00005000H

EA = EAX + EBX × 2 = 1000H + 2000H × 2 = 1000H + 4000H = 5000H

ECX = Value stored at physical address 00005000H.

Q12. In a RISC-V processor, the instruction 1d t0, 8(s1) uses which addressing mode? What is the role of s1 and 8?

A) Register Indirect; Base = s1, Offset = 8
B) PC-Relative; PC = s1, Offset = 8
C) Indexed; Index = s1, Base = 8
D) Immediate; Value = 8

Refer to the RISC-V instruction set using the internet to answer this question.

Solution:

A) Register Indirect; Base = s1, Offset = 8

Load instruction in RISC-V uses base addressing: EA = Contents of s1 + sign-extended 8-byte offset.

Q13. In an 8085 system, LDAX B uses:

A) Register Indirect
B) Direct
C) Immediate
D) Implied

Refer to Intel 8085 instruction set using the internet to answer this question.

Solution:

A) Register Indirect

LDAX B loads A from memory location pointed to by BC register pair (B=high byte, C=low byte).

Q14. Which addressing mode cannot be used for destination operands in typical ISAs?

A) Immediate
B) Register
C) Direct
D) PC-Relative

Solution:

A) Immediate

Immediate mode embeds constants in instructions. Writing to a constant is semantically invalid.

Q15. In an 8086 system, the following instructions are executed:

Assembly:

MOV AX, 1234H

MOV [2000H], AX

LES DI, [2000H]

If DS = 0100H, what are the values of ES and DI after execution?

A) ES = 1234H, DI = 0100H

B) ES = 0100H, DI = 1234H

C) ES = 1234H, DI = 5678H

D) ES = 5678H, DI = 1234H

Solution:

C) ES = 1234H, DI = 5678H

Explanation:

- `MOV [2000H], AX` stores 1234H at [DS:2000H] = (0100H × 10H) + 2000H = 03000H
- Memory layout: 03000H = 34H, 03001H = 12H, 03002H = 78H, 03003H = 56H
- `LES DI, [2000H]` loads DI from [2000H] (34H,12H → DI=1234H) and ES from [2002H] (78H,56H → ES=5678H)

Q16. Consider a microprocessor with a 20-bit address port. Suppose the opcode field takes 8 bits,then

What will be the length of an instruction if it follows (a) 4-address instruction format, (b) 2-address instruction format.

The microprocessor has a 20-bit address port, meaning each address is 20 bits long. The opcode field is 8 bits.

(a) For a 4-address instruction format, the instruction includes the opcode and four addresses. Thus, the length is calculated as follows:

Opcode: 8 bits

Addresses: 4 × 20 bits = 80 bits

Total: 8 + 80 = 88 bits

(b) For a 2-address instruction format, the instruction includes the opcode and two addresses. Thus, the length is calculated as follows:

Opcode: 8 bits

Addresses: 2 × 20 bits = 40 bits

Total: 8 + 40 = 48 bits